DEFORMATION ANALYSIS OF THE TAPERED INFLATABLE BEAM

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ABSTRACT

In the theory research and engineering practice, more basic inflatable models are essential for the mechanical property analysis of inflatable structures. Firstly, this paper presents a model of the tapered inflatable cantilever beam based on Timoshenko's theory and analyzes its deformation under a concentrated force. Moreover, the following forces resulting from internal pressure and taper ratio are introduced into the equilibrium equations of the deformed configuration. Thus, the model is optimized compared to the existing one for a straight beam. To verify the effectiveness and the superiority of the established model, the theoretical method based on the model and FEM method are compared by adopting an example about the tapered beams. Finally, the theoretical method is applied in analyzing the influence of geometry and estimating a valid range of taper ratio. By the criterion of the same amount material area, the optimum taper ratio is obtained.

Keywords: Tapered inflatable beams, Following force, Optimum taper ratio.

1. INTRODUCTION

Inflatable structures have wide applications in engineering field because they have some advantages than ordinary steel structures: high stiffness-mass ratio, small folded volume, easy transportation and installation. Especially in aerospace industry, there are airships, space antenna, solar sails, etc. The core factor in studying the inflatable structures is the internal pressure, which is the greatest difference from the ordinary solid structure.

As a basic component of inflatable structures, inflated beams get the first attention from scholars. It should be pointed out that most of research methods for inflatable beams are about straight ones. Comer and Levy [1] gave the earliest analytical expressions for the deflections and collapse loads of inflatable tubes. The Euler-Bernoulli beam's theory was used in the model. Main *et al*. [2-3] conducted the experiment of inflated tubes and compared the results with Comer's analytical results. Besides, wrinkling phenomenon and the collapse loads were analyzed by considering the bi-axial stress state. Because internal pressure doesn't appear in the deformation equations with Euler-Bernoulli beam's theory, many scholars adopted Timoshenko beam's theory to improve the results. Fichter [4] used the principle of minimum potential energy to analyze the inflatable beam and get the expressions including the internal pressure. Wielgosz and Thomas [5-6] regarded the internal pressure as the following force and obtained the effect of the internal pressure on bending and shearing stiffness by adding the force into equilibrium equations. Le van and Wielgosz [7] extended Fichter's theory by considering all nonlinear terms of kinematic and equilibrium equations. They gained the results of the inflatable tubes' bending and buckling through using principle of virtual work in the form of full Lagrange.

On the basis of previous researches, some scholars focus on geometric nonlinearity and fabric material properties to precisely describe the inflatable structures. Apedo [8] built a 3D model of inflatable beams containing geometric nonlinearity and the following effect of pressure. For inflatable beams and arches, this model could solve bending and buckling problems. William G [9] studied the arch's deformation and buckling response through establishing Timoshenko's beam element model. The results showed that the work of internal pressure done in the deformation would increase the arch bearing capacity and the shear deformation of fabric membrane would decrease. Quang-Tung and Thomas [10] divided the model into two configurations: the natural configuration (no pressure and loads) and the pre-stressed configuration (only pressure and no loads). They studied the constitutive relation changes of membrane material from the natural configuration to the pre-stressed configuration. Besides, a nonlinear system of equations was obtained and the linearization formulations showed how the internal pressure modified the bending and shear stiffness.

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DOI : 10.1017/jmech.2017.17
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One of the difficulties in studying the structural deformation problems is about the solution method. There are some analytical and numerical methods in existing literatures. S. J. Liao [11] proposed the homotopy analysis method (HAM) which could transform nonlinear buckling equation with variable coefficients into an infinite number of linear ordinary differential equations with constant coefficients. C. Cui [12] developed the method that the variable cross-section beam was modeled as a number of segments connected by continuity condition. In the model, each segment was assumed to obey characteristic of uniform cross section. Y. Q. Yang [13] studied the nonlinear vibration problems of the circular cylindrical shell and presented an improved nonlinear model which reduced the solving difficulty effectively. Besides, the author developed an averaging method to solve the second-order nonlinear partial differential equations and the harmonic balance method to solve a third-order nonlinear partial differential system [14-15].

Recently, tapered inflatable beams begin to get attention and are widely used in aerospace field, such as airships and inflatable space antenna. The airship is a variable cross-section beam, which can be divided into many tapered inflatable beams. The inflatable support tubes in space antenna can be selected as tapered inflatable beams. Compared with the straight beams, tapered inflatable beams have better mechanical characteristics and can be optimized by changing the taper ratio. Up to now, only a few results on the topic have been reported in the literature. S. L. Veldman [16] studied the bending problem of tapered inflated beam by Mohr's circle and minimum strain wrinkling criterion, and derived the initial wrinkling location and collapse moment. Also, Veldman got the optimized geometry to minimize the deflection using the same amount of skin material [17]. H. F. Tan and Z. Y. Du [18] did research on equivalent bending stiffness of tapered inflated beam according to the deflection formula of cylindrical beam. They concluded that tapered inflated beam have constant stiffness before wrinkle occurs. Although some preliminary results about collapse moment have been shown, the deformation analysis has not been intensively studied.

In this paper, we try to make the first attempt to establish an optimized bending model of the tapered inflatable beam based on the theory of straight cylindrical beams. Through analyzing the following force resulting from the taper ratio and internal pressure, the equilibrium equations are built. Because the equations belong to complicated second-order equations with variable coefficients, numerical solutions are adopted by using the Runge-Kutta method instead of analytical ones. The effects of geometric characteristics on the deformation had been accurately analyzed based on the theoretical method.

2. ANALYTICAL MODEL OF THE TAPERED INFLATABLE BEAM

This paper studies the deformation of the tapered inflatable beam under a concentrated force. Considering

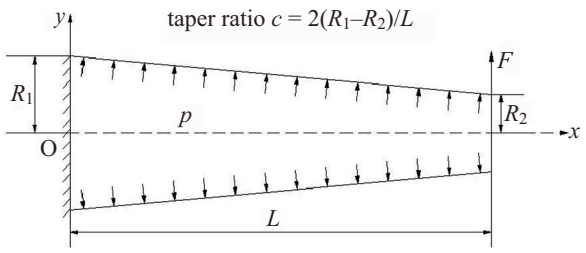

Fig. 1 The configuration of the tapered inflatable beam.

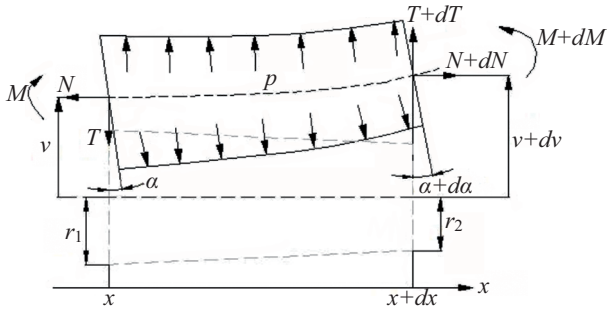

Fig. 2 The small element of the tapered inflatable beam.

transverse shear effect, the Timoshenko's beam theory is applied to the model. There are some assumptions for the model:

a. The internal pressure is high enough to keep the structure from wrinkling and breaking off.
b. The cross-section remains the same shape after deformation and is perpendicular to the central line of the beam.

As shown in Fig. 1, the tapered inflatable beam consists of the side wall and two end faces and the whole structure is closed. The structure is made of isotropic fabric membrane. Enough gas is inflated through one inflation valve into the beam to form the desired structure. L is the length of the tapered inflatable beam; R_1 is the radius of the fixed end; R_2 is the radius of the free radius; t ($t << R_1$) is the thickness of the membrane; p is the internal pressure and F is the concentrated force at the free end. The taper ratio c is defined as $c = 2(R_1–R_2)/L$.

The radius corresponding to axial position x is $r(\mathrm{x}) = \left(R_1 - \frac{c}{2}x\right)$. According to the assumption $t << R_1$, the cross-sectional area and cross-sectional moment of inertia are acquired as follows:

$$S(x) = 2\pi t\left(R_1 - \frac{c}{2}x\right), \quad I(x) = \pi t\left(R_1 - \frac{c}{2}x\right)^3 \tag{1}$$

where, $S(x)$ represents the cross-sectional area, $I(x)$ represents the cross-sectional moment of inertia.

A small element at position x is shown in Fig. 2. It defines main denotations: $N(x)$ is the axial resultant stress, $T(x)$ is the resultant shear stress, $M(x)$ is the bending moment, $v(x)$ is the deflection of the section center, $\alpha(x)$ is the angle of rotation, r_1 is the left radius of the small element, r_2 is the right radius, and p is the internal pressure.

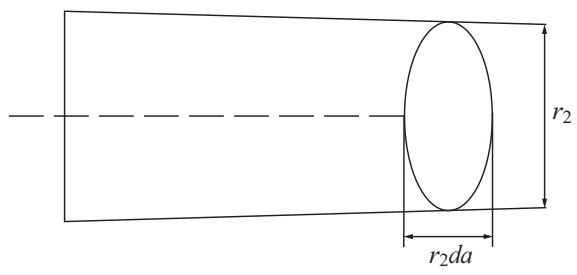

Fig. 3 Top view after deflection.

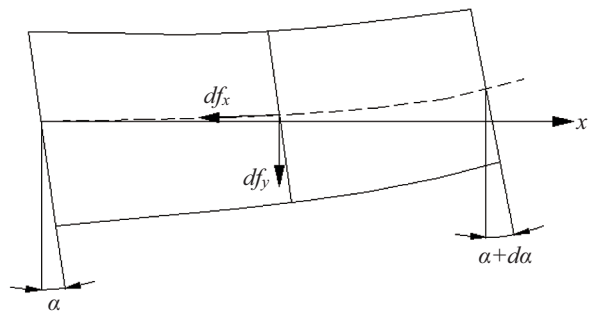

Fig. 4 Equivalent force from internal pressure.

2.1 The Effect of Internal Pressure and Taper Ratio

The small element consists of side walls and internal gas. The internal pressure is considered as part of the structure. The effect of internal pressure acting on walls can be treated as the equivalent force. In the x-direction, the equivalent force is:

$$\begin{aligned} df_x &= pS_{projection} = p\pi\left(r_1^2 - r_2^2\right) = p\pi\left(r_1 + r_2\right)\left(r_1 - r_2\right) \\ &\approx p\pi c\left(R_1 - \frac{c}{2}x\right)dx = P_1 dx \end{aligned} \tag{2}$$

where, we assume that $r_2 + r_1 \approx 2r = 2\left(R_1 - \frac{c}{2}x\right)$, $r_1 - r_2 = \frac{c}{2}dx$ based on geometry approximation. The equivalent force density of internal pressure on the side wall in the x-direction is $P_1 = p\pi c\left(R_1 - \frac{c}{2}x\right)$.

The deflection along y-direction and the rotation around z-axis occur under the load. Then the area difference between the up and down side walls appears. Then the pressure will result in additional force. As shown in Fig. 3, the projection from y-direction is an ellipse after the deflection. Therefore, the additional force in the y-direction is

$$df_y = p\pi r_2^2 d\alpha = p\pi\left(R_1 - \frac{c}{2}x\right)^2 d\alpha = P_2 d\alpha \tag{3}$$

where, we assume that $r_2 \approx r = R_1 - \frac{c}{2}x$. The equivalent force density of pressure on the side wall in the y-direction is $P_2 = p\pi\left(R_1 - \frac{c}{2}x\right)^2$.

Figure 4 shows the equivalent forces from internal pressure on the small element, that is, df_x and df_y represent the effect of internal pressure on the deflected side walls.

2.2 Equilibrium Equations

According to the model established in Fig. 2, the equilibrium equations are expressed as follows:

$$\begin{cases} \left(N + dN\right) - N - df_x + df_y \cdot \alpha = 0 \\ \left(T + dT\right) - T - df_y - df_x \cdot \alpha = 0 \\ \left(M + dM\right) - M + df_x \cdot \left(dv/2\right) - \left(N + dN\right)dv \\ \qquad + \left(T + dT\right)dx = 0 \end{cases} \tag{4}$$

The first and second formulas in Eq. (4) can be integrated as

$$N = \pi pc\left(R_1 x - \frac{c}{4}x^2\right) + C_1 \tag{5}$$

$$T = P_2\alpha + C_2 \tag{6}$$

where, C_1 and C_2 represent the integration constants.

In this paper, we assume that the constitutive relation of inflatable beam's material is isotropic. The elastic modulus of the material is E; the shear modulus is G; the shear factor K of circular section beam is 0.5 based on Cowper's theory in the reference [19]. Since the inflatable beam's shear effect must be considered, the Timoshenko beam's theory is used in the small element.

$$\begin{cases} T = KGS\left(\frac{dv}{dx} - \alpha\right) \\ M = EI\frac{d\alpha}{dx} \end{cases} \tag{7}$$

Eqs. (6) and (7) are combined and then the relationship between the deflection and rotation angle can be expressed as:

$$\frac{dv}{dx} = \left(\frac{P_2}{KGS} + 1\right)\alpha + \frac{C_2}{KGS} \tag{8}$$

Eqs. (5), (6) and (8) are substituted into the third one in the Eq. (4). Thus, the final differential equation is written as

$$\begin{aligned} &EI\frac{d^2\alpha}{dx^2} + \frac{d(EI)}{dx}\frac{d\alpha}{dx} - \left[N + \frac{N - KGS}{KGS}P_2\right]\alpha \\ &\quad + \frac{N - KGS}{KGS}\left(\pi pR_1^2\alpha(0) - C_2\right) = 0 \end{aligned} \tag{9}$$

2.3 Boundary Conditions

In order to determine the integration constants C_1 and C_2, some boundary conditions must be added. The constraint conditions are as follows:

$$\alpha(0)=0, \quad v(0)=0$$
$$\frac{dv}{dx}(0)=\frac{M(0)}{EI(0)} \tag{10}$$
$$T(0)=F, \quad N(0)=\pi p R_1^2, \quad M(0)=FL$$

Eq. (10) is substituted into Eqs. (5) and (6) and then the expressions of C_1 and C_2 can be obtained:

$$C_1=\pi p R_1^2, \quad C_2=F \tag{11}$$

2.4 Solution of the Differential Equation

After substituting the boundary conditions into the differential equation, the equation is simplified as follows:

$$\frac{d^2\alpha}{dx^2}+P(x)\frac{d\alpha}{dx}+Q(x)\alpha=f(x) \tag{12}$$

where,

$$P(x)=\frac{3c}{2(R_1-cx/2)}$$

$$Q(x)=-\frac{1}{EI}\left[N+\frac{NP_2}{KGS}-P_2\right]$$

$$f(x)=\frac{N-KGS}{EIKGS}C_2$$

The equation is a second-order differential equation with variable coefficients. It is difficult to obtain the analytical solution, so the numerical method is a good choice. Runge–Kutta method [20] is a high-precision single-step algorithm to solve the differential equations and widely applied in engineering. This paper adopts this algorithm with variable step length, which uses the fourth-order method to provide the candidate solutions and the fifth-order method to control the error. The method can satisfy the accuracy requirements.

3. VERIFICATION

For inflatable beams with fabric material, 3D membrane model is regarded as a precise simulation method. Many scholars have used 3D membrane model to verify their new theory about inflated structures. This section adopts FEM (Finite Element Method) to build a 3D membrane model of the tapered beam by the software ANSYS.

The parameters of the tapered beam are as follows: the length is 1m, the radius of the supported end is 0.0685 m and the taper ratio is 0.0755. This beam has the same amount material area with a straight beam whose has 0.05-m-radius and 1-m-length. The internal pressure p = 30000 Pa, the supported end is fully constrained, and the free one is submitted to a concentrated load F. The beam is made of fabric membrane material with the thickness 0.02 mm, Elasticity modulus 4 GPa and Poisson ratio 0.3.

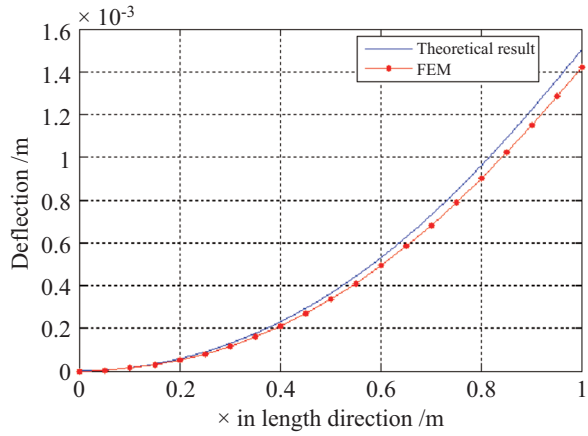

Fig. 5 Comparison of theoretical and FEM results of a tapered inflated beam, F = 2N.

The proposed theory is valid as long as no wrinkles appear in the membrane because the principal stress at any point of the membrane must be non-negative. According to the reference [21], the concentrated load F must be less than the critical wrinkling force,

$$F_w=\frac{\pi p r^3}{2L} \tag{13}$$

F_w = 5.89 N when p = 30000 Pa. Therefore, we choose F = 2N.

In the FEM model, the element type is chosen as SHELL181, which is suitable for the nonlinear large deformation and calculates the change of load stiffness caused by the pressure distribution. Besides, a suitable mesh density is important. After some attempts, the final mesh result is 200 divisions in length direction and 40 divisions in the hoop direction. For membrane structures, the internal pressure is essential to make membrane equipped with stiffness. So pressure should be applied first and make the fabric material full of stiffness. Then the concentrated force is applied at the free end, and the large deformation occurs.

The theoretical and FEM results of the tapered beam are compared in Fig. 5. The maximum difference value is 5.55% when the concentrated load is 2N. Since the equivalent force from internal pressure is calculated by the approximation method, the theoretical deflection will be a little larger than the actual value when taper ratio is not equal to zero. However, the accuracy can meet the engineering requirements and this theory about tapered beams can provide the foundation for the complex inflatable structures.

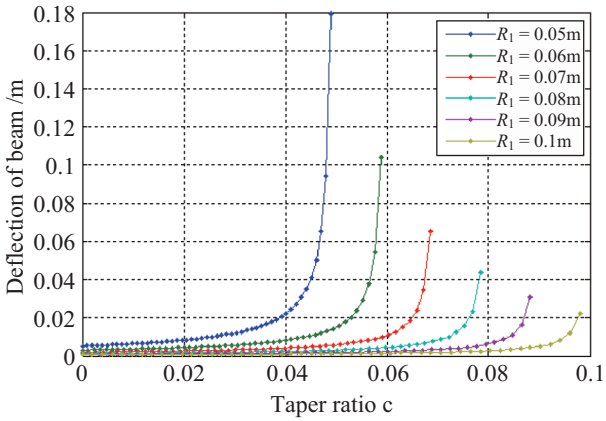

Fig. 6 Influence of taper ratio on deflection for different beams. Note: R_1 is the radius of the fixed end, L = 1 m, F = 2N.

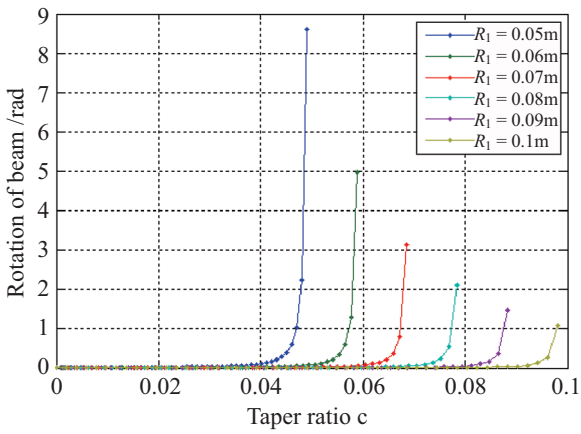

Fig. 7 Influence of taper ratio on rotation angle for different beams. Note: R_1 is the radius of the fixed end, L = 1 m, F = 2N.

4. INFLUENCE OF TAPER RATIO ON DEFORMATION

4.1 Range of Valid Taper Ratio

Tapered beams have the advantage of saving material and reducing weight. The larger the taper ratio is, the more obvious the advantages will be. However, large taper ratio will lead to a significant reduction of the beam's stiffness and even the fracture of the whole structure. Therefore, it is necessary to choose the appropriate taper ratio to ensure the stiffness of the structure. This section studies the influence of the taper ratio on deformation of the tapered inflatable beam.

Based on the theoretical solution above, this part analyzes the influence of taper ratio on the beam's deflection and rotation angle. The results are shown in Fig. 6 and Fig. 7. The parameters of the model are listed: F = 2N, p = 30000 Pa, L = 1 m, R_1 ranges from 0.05 m to 0.1 m, the taper ratio c varies from 0 to approximate $2R_1/L$.

As shown in Fig. 6 and Fig. 7, there are two stages for every curve: the first one is a slow-growth stage; the second one is a fast-growth stage after a failure point. The results show that the bending stiffness of the beam deceases rapidly when the taper ratio is greater than the critical failure value. In order to apply the results easily, an estimated range of valid taper ratio is given as follow,

$$c_{valid} \le C\frac{2R_1}{L} \tag{14}$$

where, $2R_1/L$ is the taper ratio $c = 2(R_1-R_2)/L$ when the radius of free end R_2 = 0. In other words, it is the maximum taper ratio of one beam with the radius of fixed end R_1 and length L. $C\frac{2R_1}{L}$ is a conservative estimated taper ratio at the failure point where the slope of each curve begins to increase significantly. The constant C represents the proportion of taper ratio value in $2R_1/L$ and $C \approx 0.75$.

This valid taper ratio can provide basis for the engineering design and reduce the possibility of structure failure.

4.2 Optimum Taper Ratio

Inflatable structures are widely used in the aerospace industry. However, the cost of the special material is remarkably high and light weight is one of the main goals in structural design. So it is necessary to design the optimal geometry for the minimum deflection with the same amount of material.

A numerical example is given in this section. The total amount of material is set as the surface area of a straight beam with length L = 1 m and radius R_0 = 0.05 m. The engineering demand is that the length of the beam must be 1 m. So the geometry can vary from a straight beam to a cone beam. The relationship of R_1 and c should satisfy the following constraints:

$$\begin{cases} \pi\left(R_1^2+R_2^2+R_1\sqrt{(R_1-R_2)^2+L^2}+R_2\sqrt{(R_1-R_2)^2+L^2}\right) \\ \qquad = 2\pi R_0^2+2\pi R_0 L \\ c=\dfrac{2(R_1-R_2)}{L} \\ 0\le c\le 0.75\cdot\dfrac{2R_1}{L} \end{cases} \tag{15}$$

According to Eq. (15) and the differential equation Eq. (12), the deflection at the free end of the beam can be obtained. The curve of deflection and taper ratio is shown in Fig. 8. A minimum deflection can be found at taper ratio 0.0755. Veldman S. L. [12] proposed the optimum geometry was a beam with $TR = r_{min}/r_{max}$ of approximate 0.5. The result in this paper is equivalent to TR = 0.45, which is close to the approximate value 0.5. Because the optimization criterions are a bit different, it is reasonable for the difference. Veldman S. L's criterion is that the wrinkling load should be larger than or equal to the applied load.

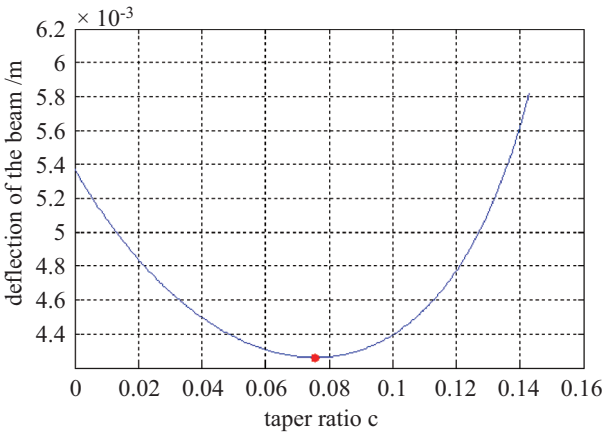

Fig. 8 Deflection of different geometry beams with the same area of material $F = 2$N, $L = 1$ m.

5. CONCLUSIONS

An analytical method to study the deformation of a tapered inflatable cantilever beam is presented in this paper. The following forces caused by internal pressure and taper ratio are introduced into the equilibrium equations of the deformed configuration. The equilibrium equation is quite complicated in solution since it is a second-order differential equation with variable coefficients. Therefore, it is a better way to use a numerical method. Commonly used Runge-Kutta method is adopted to obtain the solution. There are some meaningful results in this study shown as follows:

1. The internal pressure is considered as part of the structure. And the following forces are used to quantitatively describe the comprehensive effect of internal pressure and taper ratio.
2. The FEM and theoretical deflection results of a tapered inflatable beam are compared and agree well with each other. The accuracy of the analytical method is verified. The maximum difference is only 5.55%, which is enough for engineering requirement.
3. The influence of geometry has been analyzed and the valid range of taper ratio is less than $(0.75{\cdot}2R_1)/L$. With the same amount of material, the optimum taper ratio for a 1-m-long beam is 0.0755. These results provide the reference for engineering application of inflatable structures.

ACKNOWLEDGEMENTS

The present work was supported by State Natural Sciences Foundation "The multi-scale progressive damage analysis about the bearing layer avulsion of stratosphere airship envelope".

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(Manuscript received November 27, 2016,
accepted for publication January 26, 2017.)

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